

SUDHA LAKSHMI. R

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PROFESSIONAL SUMMARY

Computer Science undergraduate with strong software engineering fundamentals and hands-on full-stack product experience using React and Node.js. Genuinely curious about applied AI — has built and shipped an LLM-powered product (integrating Llama 3 via Groq) covering real-time inference, automated evaluation/scoring logic, and client-side ML for real-time signal detection. Comfortable reasoning about system tradeoffs across the frontend, backend, and data layer, with a solid grounding in Object-Oriented Programming, data structures, algorithms, and the probability, statistics, and linear algebra that underpin ML-driven systems. Thrives in fast-paced, collaborative environments and enjoys learning by building, experimenting, and iterating.

CORE TECHNICAL STRENGTHS

- Product Engineering: React, Node.js, Next.js — building and shipping full-stack products end to end
- Backend Systems: Node.js and Java Spring Boot, RESTful API design, JWT-based auth, PostgreSQL-backed data layers
- Applied AI: Hands-on integration of LLMs (Llama 3 via Groq) for dynamic assessment and automated scoring; client-side ML (WebAssembly) for real-time inference
- Problem-Solving: Strong logical reasoning and analytical thinking, applied through algorithm design and systematic debugging
- Object-Oriented Programming: Encapsulation, Inheritance, Polymorphism, Abstraction
- Frameworks & Tools: Spring Boot, Hibernate, Maven, JUnit, Git, IntelliJ IDEA, VS Code
- Collaboration: Effective teamwork and communication in cross-functional, high-intensity project settings

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

SRM Institute of Science and Technology

INTERNSHIPS

Software Development Job Simulation | *JP Morgan*

- Designed and implemented a real-time data feed interface using Python and Perspective to visualize live stock price data for traders, applying analytical and logical problem-solving to financial data workflows.
- Strengthened software engineering discipline in a corporate setting — version control, systematic debugging, and writing test-ready code aligned with JP Morgan's technical standards.

Frontend Development Job Simulation | *Skyscanner*

- Built a responsive, accessible travel search UI component using React and the Backpack design system, reasoning through component architecture trade-offs for a production-scale, mobile-first experience.
- Practiced component-driven development and collaborative, standards-based engineering by writing unit tests and adhering to design guidelines in a team setting.

PROJECTS

Echo — AI-Powered Technical Interview Platform | *Tech Stack: Next.js, React, PostgreSQL (Supabase), Groq (Llama 3), Google MediaPipe*

- Architected a full-stack AI product using Llama 3 for dynamic technical assessments and automated scoring, reasoning about model output evaluation to enable personalized, data-driven candidate assessment at scale.
- Engineered client-side machine learning via WebAssembly for real-time, performance-sensitive anticheat monitoring, weighing latency, accuracy, and system-design tradeoffs to keep pose and gaze detection reliable in production.

CodeCollab – Real-Time Collaborative Code Editor | *Tech Stack: React, Node.js, WebSockets, Monaco Editor*

- Developed a browser-based collaborative coding environment supporting real-time multi-user editing, implementing an Operational Transformation (OT) algorithm to resolve concurrent edit conflicts.
- Integrated syntax highlighting for 10+ languages via Monaco Editor and WebSocket-based live session sharing for seamless remote pair programming.

LANGUAGES KNOWN

Tamil | English | Hindi | Malayalam